DamRL: Reservoir Release Control with Deep Q-Learning

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CSL3090 Artificial Intelligence, Assignment 2
GitHub: https://github.com/sudochill042/DamRL_CSL3090_A2_Mini_Project

1 Problem

A reservoir operator decides every day how much water to release downstream. Releasing too little before the monsoon leaves no room for floods; releasing too much leaves shortages in the dry season. We model one year of daily operation of a single reservoir as a Markov decision process and train DQN and Double DQN agents on a synthetic simulator with stochastic inflows. Agents are compared with fixed-release, greedy and seasonal rule-curve baselines, including on drought, extreme-flood and shifted-monsoon years that the agents never saw during training. All simulator constants are assumptions, kept in `configs/default.yaml`. Units are MCM (million cubic metres) and MCM/day.

2 State Space

The observation at day t is a 10-dimensional vector, all entries scaled to roughly $[0, 1]$:

$$s_t = \big[\hat{S}_t,\ \sin\theta_t,\ \cos\theta_t,\ \tfrac{I_{t-1}}{30},\ \tfrac{\bar{I}_7}{30},\ \tfrac{D_t}{10},\ \tfrac{R_{t-1}}{30},\ \tfrac{\tilde{I}_t}{30},\ \tfrac{\tilde{I}_{t+1}}{30},\ \tfrac{\tilde{I}_{t+2}}{30}\big]$$

where $\hat{S}_t = (S_t - S_{\min})/(S_{\max} - S_{\min})$ is the storage fraction, $\theta_t = 2\pi t/365$ encodes the season, I_{t-1} is yesterday's inflow, $\bar{I}_7$ the 7-day mean inflow, D_t today's demand, R_{t-1} the previous release and $\tilde{I}_{t+k}$ a noisy inflow forecast with relative error $\sigma_k = 0.15(k+1)$. The agent never sees the true inflow of the current day. Capacity $S_{\max} = 1000$, dead storage $S_{\min} = 100$, flood threshold $S_f = 850$.

3 Action Space

Seven discrete release commands, $a \in \{0, 5, 10, 15, 20, 25, 30\}$ MCM/day. A command larger than the water available above dead storage is clipped and counted as a violation. An optional action mask removes commands that would breach dead storage even with zero inflow.

4 Reward Function

After the release the storage updates by mass balance, $S_{t+1} = \min(S_{\max},\ S_t + I_t - R_t - E_t)$ with evaporation $E_t = 0.0005\,S_t$ and spill $Q_t = \max(0,\ S_t + I_t - R_t - E_t - S_{\max})$. The daily reward is

$$r_t = -\,w_s\,\tfrac{\max(0, D_t - R_t)}{10} - w_f\,\tfrac{\max(0, S_{t+1} - S_f)}{S_{\max} - S_f} - w_d\,\tfrac{\max(0, R_t + Q_t - Q_{\text{safe}})}{R_{\max}} - w_c\,\mathbb{1}[\text{clip}] + w_p\,P_t$$

with shortage, flood-pool, downstream-flow and clipping penalties and a hydropower proxy $P_t = \frac{R_t}{R_{\max}}\big(0.5 + 0.5\,\hat{S}_t\big)$ (release times head). Weights: $w_s = 1$, $w_f = 1$, $w_d = 3$, $w_c = 0.5$, $w_p = 0.2$; $Q_{\text{safe}} = 25$, $R_{\max} = 30$. An episode is 365 days starting from storage $\sim U(450, 650)$, discount $\gamma = 0.99$.

5 Environment and Algorithm

Inflow model. Daily inflow is a seasonal mean (baseflow 4 plus a Gaussian monsoon bump of amplitude 26 centred on day 200) multiplied by lognormal noise with AR(1) persistence ($\phi = 0.7$, $\sigma = 0.5$), so wet and dry spells last several days. Demand is a cosine peaking in the dry season. Four scenarios: *ordinary*; *drought* (inflow $\times 0.6$); *extreme* (a 7-day wet spell followed by a 4-day pulse at $4\times$ inflow, random start between day 180 and 240); *shifted* (monsoon centred on day 225, noisier). Agents are trained on ordinary years (or a 60/20/20 ordinary/drought/extreme mix) and tested on a frozen set of 250 seeded episodes (100 ordinary, 50 each drought, extreme, shifted). A separate frozen validation set of 90 episodes is used for model selection and baseline tuning.

Baselines. *Random*; *fixed* release of 10 MCM/day; *greedy*, which releases the level nearest to demand and switches to maximum release once storage enters the flood pool; and a *seasonal rule curve*, the standard operating rule of real reservoirs: a target storage $S^*(t)$ is a cosine between a low and high level, and the release is $D_t + g\,(S_t - S^*(t))/R_{\max}$ rounded to the nearest level. Its four parameters (low 200, high 500, lowest on day 150, gain 1) were grid-searched on the validation set.

Algorithm. DQN with an MLP Q-network ($10 \to 128 \to 128 \to 7$, ReLU), replay buffer of 10^5 transitions, batch 128, Adam (5×10^{-4}), Huber loss, target network copied every 1000 steps, ϵ decayed linearly from 1 to 0.05 over the first 60% of training. Double DQN uses the online network to choose a' and the target network to evaluate it. Each agent is trained for 300 episodes ($\approx 1.1 \times 10^5$ steps, about 15 minutes on CPU) with three seeds; the checkpoint with the best validation return is kept. DQN was chosen because the action space is small and discrete and the problem is a standard episodic MDP, so a value-based method is a natural fit; Double DQN tests whether overestimation matters here.

6 Results

Both agents beat every baseline on every scenario (Tables 1, 2). Against the tuned rule curve, DQN improves the return by 24%, cuts flood-pool days from 11.5 to 7.6 and downstream violations from 5.5 to 2.3 days per year, with 14% less total shortage. The learning curves (Fig. 1) rise past the rule curve after about 150 episodes and the three seeds agree within a few units of return. Double DQN is slightly better than DQN (-36.2 vs -37.9) but the confidence intervals overlap, so with only seven actions overestimation is not a major issue

Table 1: Mean per-episode metrics over 250 test years (agents: mean of seeds). CI is the 95% interval of the return; Short. is total shortage (MCM); Flood and Clip are days per year.

Method	Return ±CI	Short.	Spill	Flood	Clip
Random	-196.4 ± 4.6	1058	172	30.1	127.9
Fixed 10	-143.3 ± 4.5	447	457	70.2	82.9
Greedy	-102.1 ± 3.7	543	76	30.6	61.7
Rule curve	-49.9 ± 3.6	531	43	11.5	2.1
DQN	-37.9 ± 2.0	456	41	7.6	2.3
Double DQN	-36.2 ± 1.9	447	38	6.7	1.3
DQN mixed	-36.5 ± 2.2	451	43	7.7	0.1
DQN masked	-40.3 ± 3.6	506	32	5.8	0.0

Table 2: Mean test return per scenario. Agents were trained on ordinary years only (except DQN mixed).

Method	Ordinary	Drought	Extreme	Shifted
Rule curve	−29.5	−78.9	−57.9	−53.5
DQN	−17.7	−71.7	−40.0	−42.6
Double DQN	−17.5	−69.4	−37.0	−39.4
DQN mixed	−19.3	−60.9	−42.2	−40.9
DQN masked	−19.9	−82.3	−37.5	−42.0

here. The agents trained only on ordinary years still transfer to drought, extreme and shifted years, mainly because the state includes the storage fraction, season and short forecast, which is enough to react to unusual inflow. Training on the 60/20/20 scenario mix (DQN mixed, 2 seeds) gives the same overall return but improves drought years the most (−60.9 vs −71.7) at a small cost on ordinary and extreme years, so seeing dry years in training teaches the agent to save water. Action masking (1 seed) removes clipping entirely and gives the fewest flood days, but the masked agent holds back too much water, has the largest shortage and the lowest return of the agents (−40.3), so the clipping penalty alone is a better teacher than a hard mask here. Fig. 2 shows the learned behaviour: the agent runs the reservoir close to dead storage before the monsoon and delays filling during the early monsoon, which the rule curve cannot do because it tracks a fixed target, so it has more room when the flood pulse arrives.

Failure cases. Four of the five worst DQN test episodes are extreme years with 470 to 910 MCM of spill and 35 to 57 flood-pool days: the flood pulse (over 100 MCM/day for several days) arrives when storage is already high, and the 30 MCM/day outlet cannot drain the reservoir fast enough, so spill is physically unavoidable and every method suffers. The fifth is a drought year in which the agent reaches dead storage and records 241 shortage days. The hydropower term rewards release, so the agent has little reason to save water for a drought it is not told is coming.

7 Conclusion

A DQN agent with a compact state (storage, season, recent and forecast inflow, demand) learns a release policy that outperforms a tuned seasonal rule curve by about 25% in return and halves flood-pool days, and it generalises to inflow regimes it never trained on. Double DQN gives a small, not significant, further gain. Limitations: the simulator is synthetic, hydropower is a crude proxy, and the agent cannot avoid spill when a large pulse hits a full reservoir or shortage in a long drought. Natural next steps are a longer-horizon forecast, a multi-reservoir cascade and a constrained-RL formulation with hard flood limits.

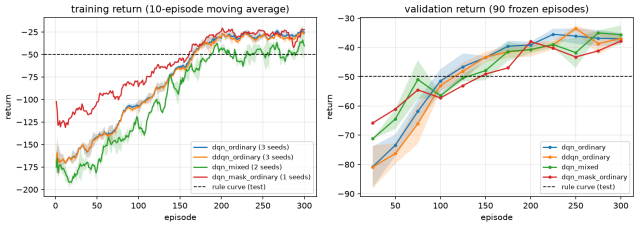


Figure 1: Training return (10-episode moving average) and validation return, mean ± std over seeds. Dashed line: rule curve on the test set.

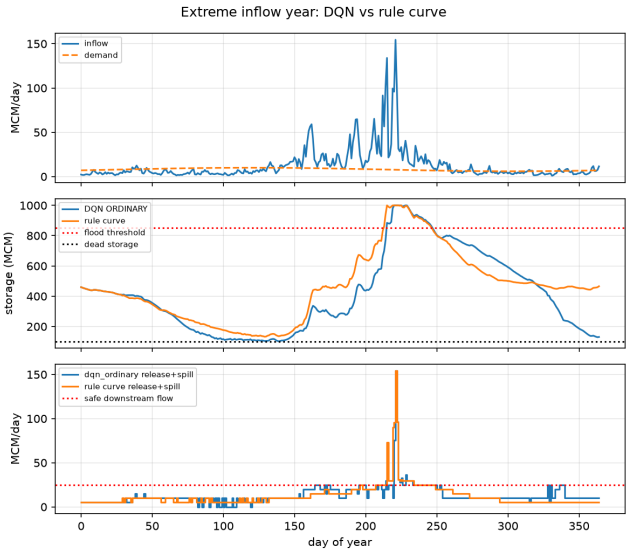


Figure 2: One extreme test year. DQN keeps storage lower through the early monsoon (days 160 to 215) so the reservoir fills later and spills less; both methods spill during the 150 MCM/day pulse. The drawdown after day 300 is a finite-horizon effect: water has no value after day 365.